

External Protection Service in Parquet

Modular Encryption - Testing Plan

Background

We expanded Arrow's Parquet Modular Encryption (PME) capabilities by allowing third-party encryption services (which may be external) to integrate into Arrow.

Our changes touched the following Arrow C++ code:

- Adding a new encryption algorithm type to the ParquetCipher struct to support the external encryption and decryption.
- File encryption and decryption properties (extended both the encryption/decryption configurations and the file encryption/decryption properties).
- Exposure of the new properties in Python bindings.
- Encryptor and decryptor class refactor.
- Added a value to the encryption algorithm type enum.
- Encryptor and decryptor creation.
- New classes for encryptors and decryptors (including new logic for performing encryption/decryption and interaction with external encryptors and decryptors).
- Adding new metadata to the Parquet file (including per-column level metadata).
- File encryption and decryption.

This document details the testing methodology employed to ensure the solution is fully functional and complies with Apache Arrow's standards. This section focuses specifically on the testing performed within the Arrow framework. The new code implementing the external encryptor and decryptor is tested independently within its dedicated repository.

These tests also include the Cython bindings which are key to exposing the Arrow changes to the application developers.

Test Focus Areas

Our focus in testing is centered around the following principles:

1. Validate the successful operation of the major refactor of encryptors and decryptors.
2. Verify that information passed within the new external objects for encryption and decryption flows correctly through the architecture, reaching the intended code locations.
3. Confirm the correct functionality of the new external encryptors and decryptors, including handling of the new ParquetCipher value.
4. Ensure the newly added metadata is accurately written and read.
5. Validate successful round-trip encryption and decryption for the specified combinations.
6. Confirm the successful application of different encryption algorithms to distinct columns.
7. Ensure the contract between Arrow and the new external agent is fulfilled.

We refer to these focus areas in the test tables below so we may guarantee that they are sufficiently covered.

Test Coverage and Focus Areas Tested

We present a list of the existing tests in Arrow and PyArrow, with a brief description of the features tested in it, as well as what was added. We relate each change in the test back to the focus features listed above to show how we achieved test coverage.

Arrow

Test: [file_parquet_encryption_test](#)

Original Scope: Unit tests for a higher level encryption test that integrates KMS use, and dataset table preparation. Looks closer to an integration test of all the encryption related functionality (including KMS). Additionally, includes a test for processing encrypted datasets over 2^{15} rows in multi-threaded mode, testing the scale of the input dataset.

Extended Scope: Add test cases for when the dataset is encrypted and decrypted using external and per-column encryption.

Focus Areas: 2, 3, 5, 6

Test: [column_writer_test](#)

Original Scope: Test encryption and decryption at the column level by specifying the column and file encryption/decryption properties directly.

Extended Scope: Add round trip encryption/decryption of external and per-column encryption using the external properties directly. Also add tests that check the metadata that was added at the per-column level is successfully written.

Focus Areas: 1, 2, 3, 4, 5, 6

Test: [types_test](#)

Original Scope: Test conversions between types and their string versions, or types and their converted types.

Extended Scope: Add an explicit check for the supported ParquetCipher types which now includes a new type for the external algorithm.

Focus Areas: 3

Test: [file_deserialize_test](#)

Original Scope: Tests that file structure and metadata is preserved on deserialization and decryption.

Extended Scope: Add test cases for the new decryption call introduced by the refactor, add a specific test that fixes a pre-existing Arrow bug that would not correctly write out the encryption algorithm used if it was different than AES_GCM_V1.

Focus Areas: 1, 2, 3, 4, 5

Test: [aes_encryption_test](#)

Original Scope: Renamed from encryption_internal_test. Unit tests that perform round trip AES encryption and decryption.

Extended Scope: Ensure backwards compatibility with AES encryption and decryption after the refactor.

Focus Areas: 1

Test: [crypto_factory_test](#)

Original Scope: New test added.

Extended Scope: Test conversion of all encryption and decryption configurations into their file properties counterparts. Includes the original Arrow versions and the external versions added for this project.

Focus Areas: 1, 2, 3

Test: [properties_test](#)

Original Scope: Unit tests for creating different combinations of encryption and decryption properties.

Extended Scope: Add tests for when the encryption algorithm is specified per-column, making sure the properties persist. Test the inheritance of the added external classes with respect to their original Arrow counterparts. Test that the per-column encryption algorithm is respected.

Focus Areas: 1, 2, 3, 4, 6

Test: [external_dbpa_encryption_test](#)

Original Scope: New test added.

Extended Scope: Comprehensive test of round trip encryption and decryption using the external DBPA interface. Includes testing metadata, keys, and per-column encryption.

Focus Areas: 2, 3, 4, 5, 6, 7

Test: [per_column_encryption_test](#)

Original Scope: New test added.

Extended Scope: Explicitly test round trip encryption and decryption using different algorithms per column.

Focus Areas: 1, 2, 3, 4, 5, 6

Test: [encoding_properties_test](#)

Original Scope: New test added.

Extended Scope: Tests for ensuring the new EncodingProperties that are exchanged between Arrow and DBPA contain all necessary and correct metadata for DBPA to work.

Focus Areas: 3, 4, 7

PyArrow

Test: [test_dataset_encryption](#)

Original Scope: Test encryption and decryption of datasets that use ParquetFragmentScanOptions to read data.

Extended Scope: Add test cases for when the dataset is encrypted and decrypted using external and per-column encryption.

Focus Areas: 2, 3, 5, 6

Test: [test_external_encryption](#)

Original Scope: New test added.

Extended Scope: Comprehensive application level test that uses external and per-column encryption on data. Includes creating valid and invalid versions of the configurations.

Focus Areas: 2, 3, 5, 6

Arrow and DBPA Contract Fulfillment

A series of new tests were added that ensure the contract between Arrow and the external DBPA agent is fulfilled.

Test: [dbpa_enum_utils_test](#)

Original Scope: New test added.

Extended Scope: Test all conversions between Arrow enums and their DBPA counterparts.

Focus Areas: 7

Test: [dbpa_executor_test](#)

Original Scope: New test added.

Extended Scope: Main test for the interaction between Arrow and the DBPA Agent, ensuring encryption and decryption results are correctly sent, initialization scenarios, and bubbling up of exceptions.

Focus Areas: 7

Test: [dbpa_library_wrapper_test](#)

Original Scope: New test added.

Extended Scope: Test the proper operation of the dbpa_wrapper decorator that ensures that instances of DBPA Interface are properly destroyed, and its resources are released.

Focus Areas: 7

Test: [dbpa_test_agent](#)

Original Scope: New test added.

Extended Scope: Implementation of the DBPA Interface for testing purposes.

Focus Areas: 7

Test: [loadable_encryptor_utils_test](#)

Original Scope: New test added.

Extended Scope: Tests for loading the external agent library wrappers.

Focus Areas: 7

Manual Correctness Testing

The PME extension project focused on developing PyArrow bindings. We created several manual scripts outside the test suites to demonstrate the use of the new PME extensions, perform sanity checks, and manually verify the solution's correctness.

Test: [base_app](#)

Original Scope: New script added.

Extended Scope: Script used for round trip encryption and decryption of a small test table using external encryption and per-column encryption.

Focus Areas: 2, 3, 5, 6

Test: [use_external_dbpa_encryption](#)

Original Scope: New script added.

Extended Scope: Intended as a tutorial and guide for application developers to learn how to use external and per-column encryption. Heavily documented and with different usage examples.

Focus Areas: 2, 3, 5, 6

Test: [verify_external_roundtrip_correctness](#)

Original Scope: New script added.

Extended Scope: Comprehensive encrypt/decrypt roundtrip correctness verification for DBPA encryption. Combinatorial over data types, encoding, compression, and page versions.

Focus Areas: 2, 3, 5, 6

Instructions for Running Arrow and Python Unit Tests

Please look at the [Developing Arrow + DBPS guide document](#) for detailed instructions on setting up the development environment.

In particular, [Section 3](#) has the specific instructions needed to run the C++ and Python tests.